

Rainfall prediction by using ANFIS times series technique in South Tangerang, Indonesia

Wayan Suparta^{a, b, *}, Azizan Abu Samah^a

^a Institute of Ocean and Earth Sciences (IOES), University of Malaya, 50603, Kuala Lumpur, Malaysia

^b Department of Electrical Engineering, Institut Teknologi Nasional Yogyakarta, Jl. Babarsari, Caturtunggal, Depok, Sleman, Yogyakarta, 55281, Indonesia

ARTICLE INFO

Article history:

Received 24 February 2020

Accepted 25 August 2020

Available online 9 October 2020

Keywords:

ANFIS

Time series

Rainfall

South Tangerang

ABSTRACT

Excessive rainfall is one of the triggers for the flooding phenomenon, especially in the tropics with flat or concave areas. Some critical points in the South Tangerang region, which are currently one of the most rapidly developing cities, cannot be ignored from the flooding problem. Floods cause disturbing human activities, loss of life and property, and in turn affect the economic stretch in an area. This paper aimed to predict rainfall by exploring the application of artificial intelligence techniques such as ANFIS (Adaptive NeuroFuzzy Inference System). The proposed technique combines neural network learning abilities with transparent linguistic representations of fuzzy systems. The ANFIS model with various input structures and membership functions was built, trained, and tested to evaluate the capability of a model. Analyses of six-year rainfall data on a monthly basis in South Tangerang City, Banten found that rainfall prediction based on ANFIS time series is promising where 80% of data testing is well predicted.

© 2020 Institute of Seismology, China Earthquake Administration, etc. Production and hosting by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction

Indonesia is a country with a disaster-prone zone, starting from floods, landslides, earthquakes to tsunamis. With a remarkable tropical climate where the country tends to be hot, the temperature is not the main parameter of climate in this country. In other words, rainfall plays an important role in climate change, especially during the rainy season, where rainfall occurs quite high in each year. The rainy season usually triggers frequent of a natural disaster such as landslides and floods. Floods can be caused by various factors such as the condition of the catchment area [1], rain duration and intensity [2], land cover [3], topographic conditions [4], and drainage network capacity [5]. While the basic factor of a disaster is the

result of climate change. Among the types of floods that are very upsetting besides flash floods are urban floods.

Based on flood forecasts in Indonesia where December, January, and February throughout the year are potentially high flood months. Similar to neighboring areas like Jakarta, South Tangerang is also no exception to this flooding phenomenon. The South Tangerang City is located in the eastern part of Banten Province or borders with South Jakarta about 5 km at coordinates of 106°38'–106°47'E and 06°13'30'–06°22'30"S. Most of this area is lowland and has a relatively flat topography with a slope of 0–3% (average) while the altitude of the region is between 0 and 25 m above sea level. Looked from the climate condition, the intensity of the sun is around 50% and the highest rainfall (December–January) is reaching until 265 mm. Until 2019, there are 13 locations are regularly subscribed to flooding because of inadequate drainage [6].

With the above climatic and topographic conditions, the development of an early warning system (EWS) that can provide fast, accurate, targeted, easily accepted, and understood, reliable and sustainable information is indispensable. The previous method was to measure the height of the water level in the river and then classified according to the level of danger [7]. However, new flooding can be predicted if the river's water level is at an almost overflowing height. This causes the warning given too short and the community does not have time to prepare self-preservation. The

* Corresponding author. Institut Teknologi Nasional Yogyakarta (ITNY), Yogyakarta, 55281, Indonesia.

E-mail address: drwaynesparta@gmail.com (W. Suparta).

Peer review under responsibility of Institute of Seismology, China Earthquake Administration.



Production and Hosting by Elsevier on behalf of KeAi

existing monitoring system in this region cannot be recorded automatically and several long-term historical data cannot be utilized for the benefit of future investigations. In this study, a soft-computing method will be applied with consideration of low costs, easy to analyze and still have good accuracy. One form of hybrid softcomputing technique to be developed is the Adaptive Neuro Fuzzy Inference System (ANFIS) model. ANFIS is a combination of fuzzy logic and artificial neural networks. ANFIS model has been proposed for modeling hydrological time series [8], to develop a forecasting seasonal rainfall in Victoria of southern Australia to anticipate floods [9] and for spatial flood predictions such as Hong et al. [10] and Bui et al. [11]. Suparta and Alhasa [12–14] and Suparta and Putro [15] have applied the ANFIS to predict zenith tropospheric delays (ZTD) instead of GPS-derived ZTD to estimate atmospheric water vapour content. Previously, artificial neural networks (ANNs) technique have been extensively used as a tool to model rainfall forecasting such as combined with a linear multiple-input single-output (MISO) for daily rainfall-runoff modeling [16]. However, the power of ANNs still has a number of limitations, such as gradient descent technique might get trapped in local minima and difficulties in selecting an architectural model. Determining the optimal value of hidden layers in pure ANN depending on system behavior is a big challenge to get a local minimum. Compared to the ANN, the ANFIS model is more transparent to the user and causes less memorization errors [17]. Based on some previous investigations which indicated that ANFIS can be used as an efficient tool for rainfall modeling, it is expected that using the ANFIS network exploration in this study to predict rainfall for the particular region and condition of the process can overcome these limitations and increase their reliability and accuracy.

2. Methods

2.1. Dataset and location

A number of measurement data is collected to support the study. The primary data are obtained from MCGA (Meteorology, Climatology, and Geophysics Agency) or BMKG (*Badan Meteorologi, Klimatologi dan Geofisika*). The data processed for this study is rainfall and temperature. Since the BMKG in one district only managed one weather station, this study will use data in South Tangerang City which located in Pondok Betung (geographically: 6°15'40.22"S, 106°44'56.26"E). Data provided by BMKG is available on a daily basis and ten years of data will be employed. The location of the meteorology station is depicted in Fig. 1 where the distance from the Universitas Pembangunan Jaya (UPJ) is about 6 km.

2.2. ANFIS model development for rainfall predictions

As has been introduced, the approach used for rainfall prediction is to explore the ANFIS method. The ANFIS model applies the first order of Tagaki-Sugeno-Kang (TSK) [18,19]. From the various types of supervised learning algorithms, the hybrid learning algorithm is chosen as a rainfall prediction. The reason to choose a hybrid learning algorithm is it has been widely applied. One of the advantages of ANFIS is, it is a combination of ANN and fuzzy systems using ANN learning capabilities to obtain fuzzy if-then rules with appropriate membership functions, which can learn something from the imprecise data that has inputted and leads to the inference. Another advantage, it can make effective use of self-learning and memory abilities of neural networks and brings a more stable training process [20].

In general, ANFIS is constructed with five layers; the input layer is as antecedent parameters, three hidden layers so-called a rule-based with three constant parameters and one consequent parameter, and

one output layer (see Fig. 2). The output in each layer is expressed with the equation as presented at the bottom of the figure. In the first layer, it transforms an input into a degree between 0 and 1 (fuzzification) which is so-called premise parameter. It is an activation function with membership function such as gauss, triangular, trapezoidal, or generalized Bell. The second layer will estimate each incoming signal to each neuron using the product operator; in the third layer, it will normalize all input signals while the fourth layer will do fuzzification. Finally, in the last layer, it will summarize all the weighted output values. For more detail about the ANFIS concept, readers can refer to Suparta and Alhasa [12–15].

Fig. 2 shows the ANFIS architecture used in this work with three-time series inputs " $x(t)$, $x(t-1)$, and $x(t-2)$ " and one output of $s(t)$. The inputs are average monthly rainfall values applied to the ANFIS model to monthly rainfall prediction $s(t)$. The structure of the ANFIS model was set for three layered models which are having input, output, and hidden layer. The hidden layer is used the gbellmf and trapmf for the rainfall prediction method. The input–output structure of the prediction model can be formulated through the equation below [21]:

$$\text{Input: } w(t) = [x(t), x(t-1), x(t-2)] \quad (1a)$$

$$\text{Output: } s(t) = [X(t+1)] \quad (1b)$$

where $x(t)$ is the time in a day, $x(t-1)$ and $x(t-2)$ are prior time spans, input ($w(t)$) is a parameter designed as input for training, and output $s(t)$ is the output for predicted parameter.

In this study, meteorological parameters that were selected will be trained and tested to predict rainfall. The size of the data being trained is 60% for training and 40% for testing, or data for training is more than data for testing. The output of the system will be tested to match the target given to the system. If not, then the learning algorithm will be improved by providing new network architecture or learning. The RMSE (Root Mean Squared Error) [22] as in equation (2) is used for testing. If RSME is small enough, ANFIS is considered to have succeeded in carrying out the training process.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{k=1}^n (y_{\text{pred}} - y_{\text{obs}})^2} \quad (2)$$

The results of rainfall prediction through the ANFIS method are also evaluated with Mean Squared Error (MSE). Mathematically, MSE is expressed as [23,24].

$$\text{MSE} = \frac{1}{n} \sum_{k=1}^n (y_{\text{pred}} - y_{\text{obs}})^2 \quad (3)$$

From the observation (training and testing), input parameters in the form of meteorological data that have undergone a learning process, and the input–output structure of the performance of the networks can be formulated through the correlation coefficient (R). These parameters are often defined in terms of the prediction of error which is the difference between the actual and predicted values [25]:

$$R = \sqrt{1 - \frac{\sum_{k=1}^n (y_{\text{obs}} - y_{\text{pred}})^2}{\sum_{k=1}^n (y_{\text{pred}})^2}} \quad (4)$$

The ANFIS model is considered to be able to make predictions if the MAPE (Mean Absolute Percentage Error) value in the testing or validation process is below 20%. MAPE values can be calculated

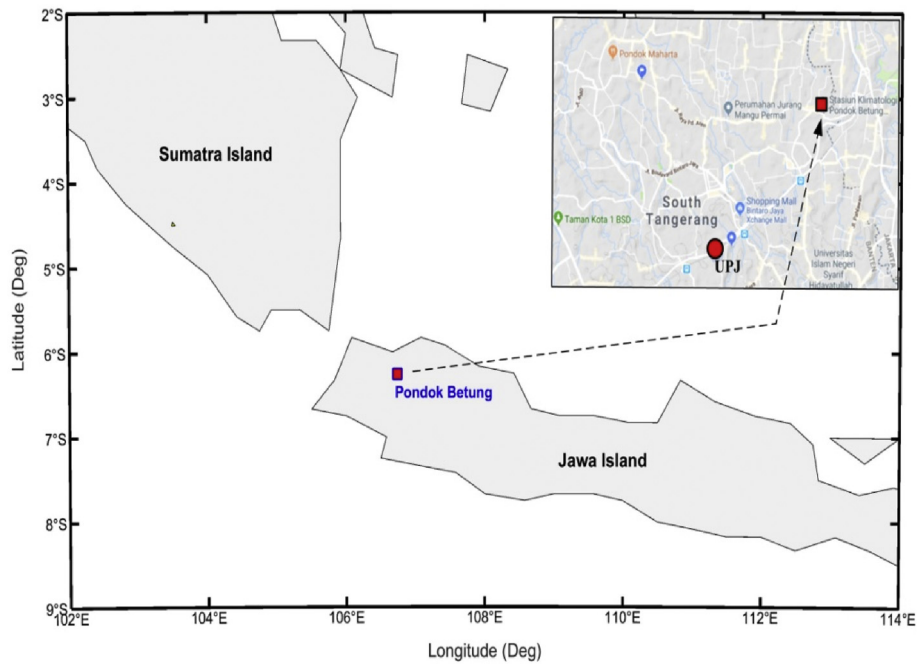


Fig. 1. The location of the study by Google Maps in South Tangerang City.

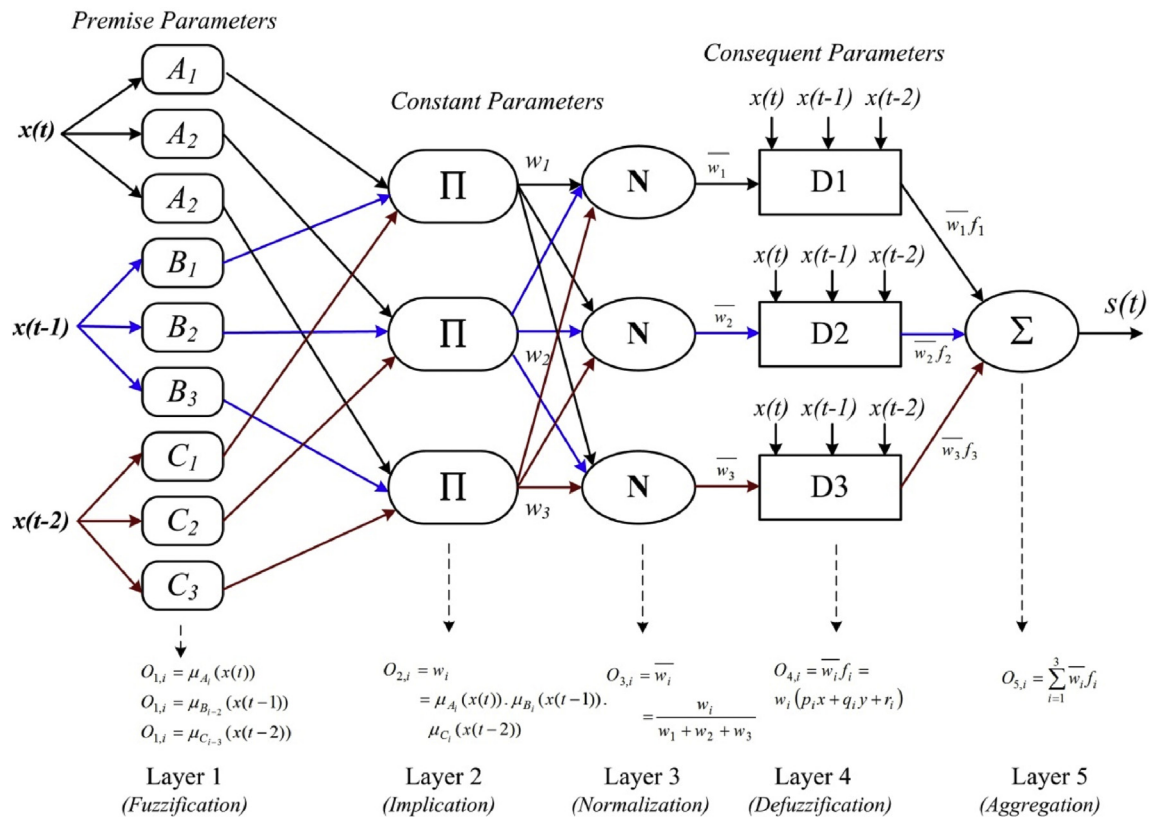


Fig. 2. ANFIS architecture with three inputs, one output, and three rules.

using the following Equation (4) [26,27] after the observation data are normalized.

$$\text{MAPE} = \frac{1}{n} \sum_{k=1}^n \left(\frac{y_{\text{pred}}(k) - y_{\text{obs}}(k)}{y_{\text{pred}}(k)} \times 100\% \right) \quad (5)$$

where y_{obs} is the value of the observations, y_{pred} is the value of the predicted results, and n is the data number.

3. Results and discussion

3.1. Meteorological parameters

Fig. 3 shows the variations of both daily and monthly average of rainfall in South Tangerang for ten-year period (2009–2018). As can be seen from the figure, in the beginning, and at the end of the year, rainfall patterns showed a higher rainfall than in the middle of the year and reached more than 75 mm. Unfortunately, rainfall during the half of 2015 and 2016 was not well recorded. Because rainfall data at certain stations also experience maintenance problems, this is the best way to estimate them based on available historical data.

3.2. Rainfall prediction

Daily data for ten years provided in Fig. 3 cannot be realized to predict daily rainfall due to very high computation problems. Instead, monthly averages are processed and then presented in a matrix size of 10×12 where 10 indicates the year of data used and 12 is the number of months. Furthermore, data in this matrix is divided into data training and data testing. To reduce computations that are too large in the pre-processing, the data is normalized into the range 0–1 using the following equation [27,28]:

$$X' = \frac{X - b}{a - b} \quad (6)$$

where X' is the normalized data, X is the original data, a and b are maximum and minimum values of original data, respectively.

In this prediction, a 10-year data is used where the training data used is rainfall data from January 2009 to June 2014 (6 years). On

the other hand, the training target is rainfall data from July 2009 to December 2014 (6 years). Training is conducted using a membership function such as trapezoid membership function (trapmf) and generalized Bell membership function (gbellmf). In this step, a fuzzy inference system (FIS) is generated and evaluated, which can produce MSE and MAPE. The rest of 4-year data (July 2015 to December 2018), it will be used for testing targets. In other words, training and testing data, in this case, by trial and error is 60%:40%. This comparison is the best optimization result for typical data obtained at BMKG station in South Tangerang. Fig. 4 shows the result of training and testing for 10-year data. The training result showed that the target and original data is closely correlated where their difference is about 0.0029%. The testing results are evaluated by varying the epoch from 10 to 50 s to show which epoch will get optimal results. However, testing targets with original data were not well predicted (no correlation at all) where the difference was about 28% and 14% on average for trapmf and gbellmf, respectively. The problem is appeared due to no data for January 2016, June 2016, and July of 2015. This implies that there is no predicted data around the missing data (see the testing results of Fig. 4). There is no recorded data at this station, possibly due to an instrumentation system that is in the process of maintenance or the system is not functioning. Incomplete data beforehand will be a serious problem to obtain accurate prediction results using ANFIS time series techniques. The missing data at any given time can be solved by interpolation, but this is not the purpose of this study.

Alternatively, to get accurate prediction results with optimum epochs, completed monthly data for a period of six years period (2009–2014) is employed. In this case, data comparison for training and testing is 3:2, which means three years of data for training (2009–2011) and two-year data for data testing (2012–2013). Fig. 5 shows the results for training and testing where the training target is a similar trend with the original data. This shows the success of ANFIS in capturing the learning process that will be used as output in the forecasting process. The variation of the epoch number from 10 to 50 s shows that the ANFIS model is capable to learn the training data to become their output. Assuming that gbell is the best of membership function, it is shown that at the epoch = 30 s the prediction results is starts in the same peak where the difference between the original target with the testing target is the

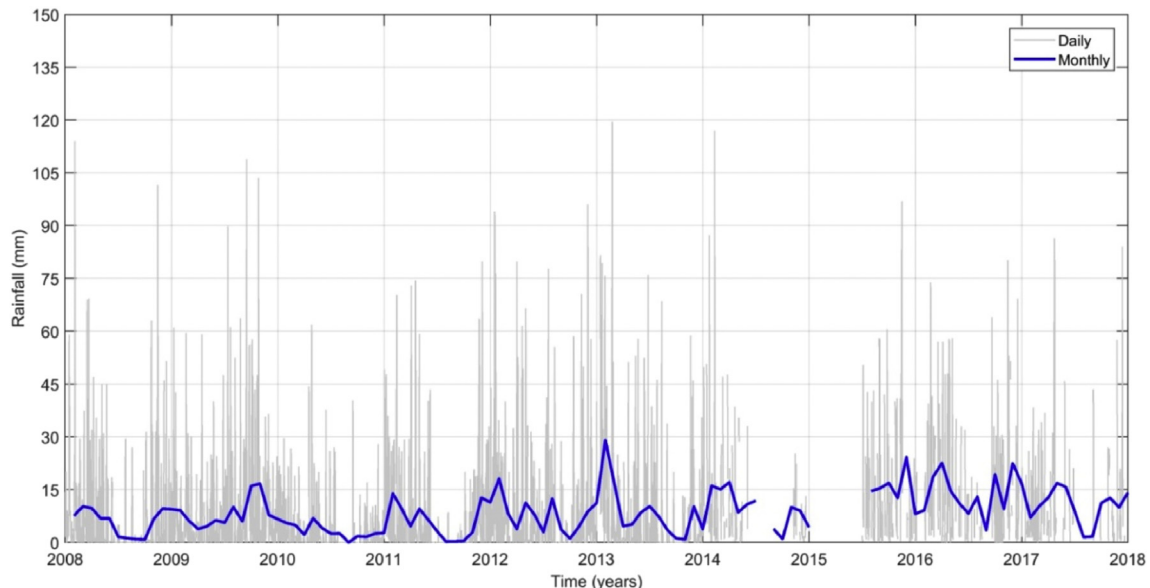


Fig. 3. Daily and monthly average variations of rainfall in South Tangerang for ten years period (2009–2018).

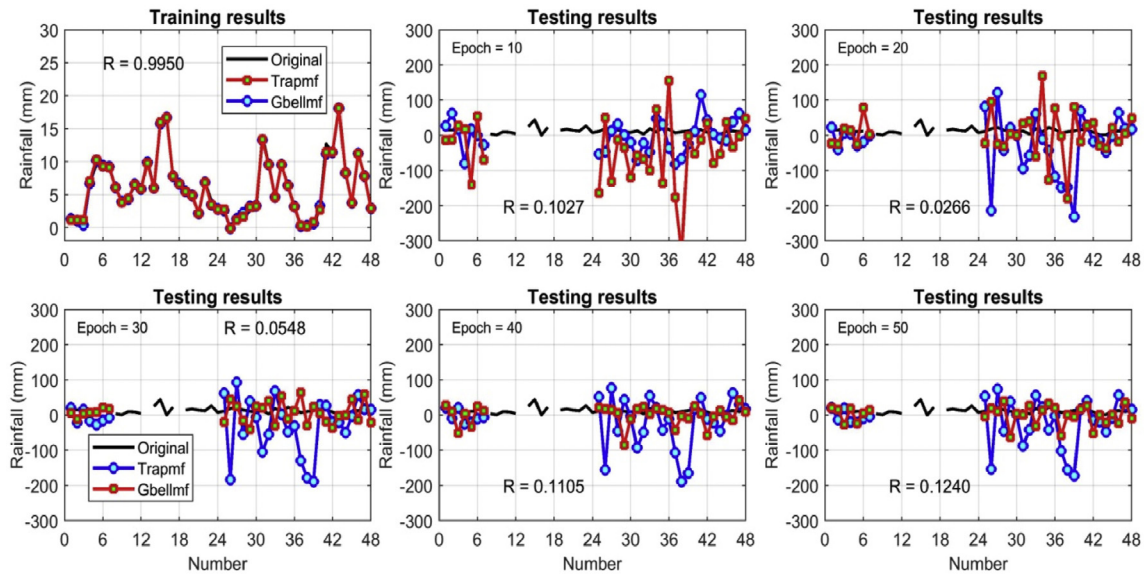


Fig. 4. The result based on ten-year data (2009–2018) to predict rainfall in South Tangerang by using (a) trapmf and (b) gbellmf where data comparison for training and testing is 6:4.

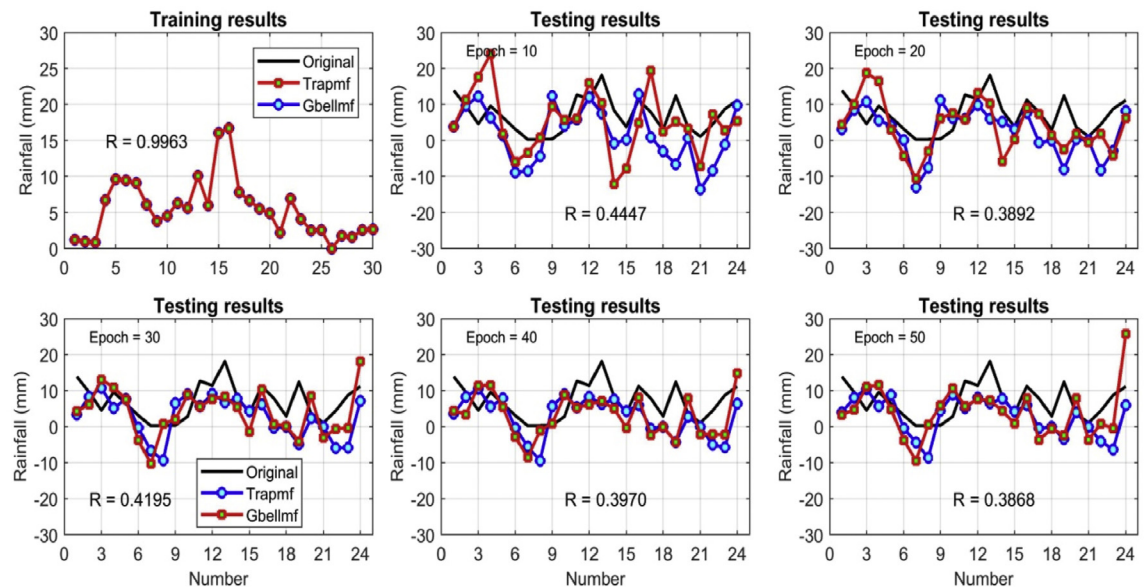


Fig. 5. Rainfall prediction results in South Tangerang with epoch (t) variations based on six-year data (2009–2014) where the comparison of data for training and testing is 3:2.

lowest (19.41%) and MAPE is also the lowest (see the performance indicator as in Table 1).

Another rainfall prediction using a data comparison between training and testing of 4:2 is shown in Fig. 6. Table 2 shows the prediction result for rainfall using gbellmf where time epoch (t) is varied from $t = 10$ to $t = 100$ s. It is found that more number of epochs, the process will require more computational time. More time for computation means not efficient and does not guarantee to get more accurate prediction results. The advantage results from Table 2 which using a comparison of 4:2 is that the MAPE testing for gbellmf is drastically reduced as compared to Table 1 which using 3:2. The RMSE in Table 2 also decreases with increasing computational time. On the other hand, the evaluation of two mapping functions found that gbellmf as illustrated in Table 1 yielded the best results.

From Figs. 5 and 6, it can be noted that the rainfall prediction has used two numbers of membership functions. As in Tables 1 and 2, MSE for testing and training is constant over epoch variations while MAPE, MSE, and RMSE tend to decrease with increasing epoch number after the optimal value. Another predictive performance as shown by the result of testing differences between original data and targets is that the smaller value of prediction differences will increase the correlation coefficient (R). An increase in the value of R indicates more accurate prediction results. On the other hand, the ANFIS architecture generated from completed data in this work, i.e. six-year data (2009–2014) has worked with 161 nodes, 448 linear parameters, 496 parameters, 30 training data pairs, and 64 fuzzy rules. At this stage, ANFIS training is completed at epoch 50. It can be seen from Table 2 that the RMSE has increased again at $t = 60$ s and decreased again until

Table 1

Summary performance for training and testing targets with epoch (*t*) variations where the comparison of training data and testing data used are 3:2.

Performance Indicator	Generalized Membership Function (gbellmf)				
	Epoch = 10	Epoch = 20	Epoch = 30	Epoch = 40	Epoch = 50
MAPE Testing	38.6752	32.8519	31.0188	32.1486	35.1420
MAPE Training	0.0052	0.0024	0.0015	0.0009	0.00085
MSE Testing	0.0865	0.0865	0.0865	0.0865	0.0865
MSE Training	0.0550	0.0550	0.0550	0.0550	0.0550
RSME	8.5095	7.4792	6.7554	6.9504	7.8422
Difference Testing	0.2421	0.2056	0.1941	0.2012	0.2199
Trapezoid Membership Function (trapmf)					
MAPE Testing	1.6332	1.4723	1.3596	1.3468	1.3072
MAPE Training	3.208×10^{-5}	1.475×10^{-5}	1.91×10^{-6}	1.637×10^{-6}	1.67×10^{-6}
MSE Testing	0.0865	0.0865	0.0865	0.0865	0.0865
MSE Training	0.0550	0.0550	0.0550	0.0550	0.0550
RMSE	8.6150	8.1394	7.4119	7.2994	7.0975
Difference Testing	0.2453	0.2211	0.2042	0.2023	0.1964

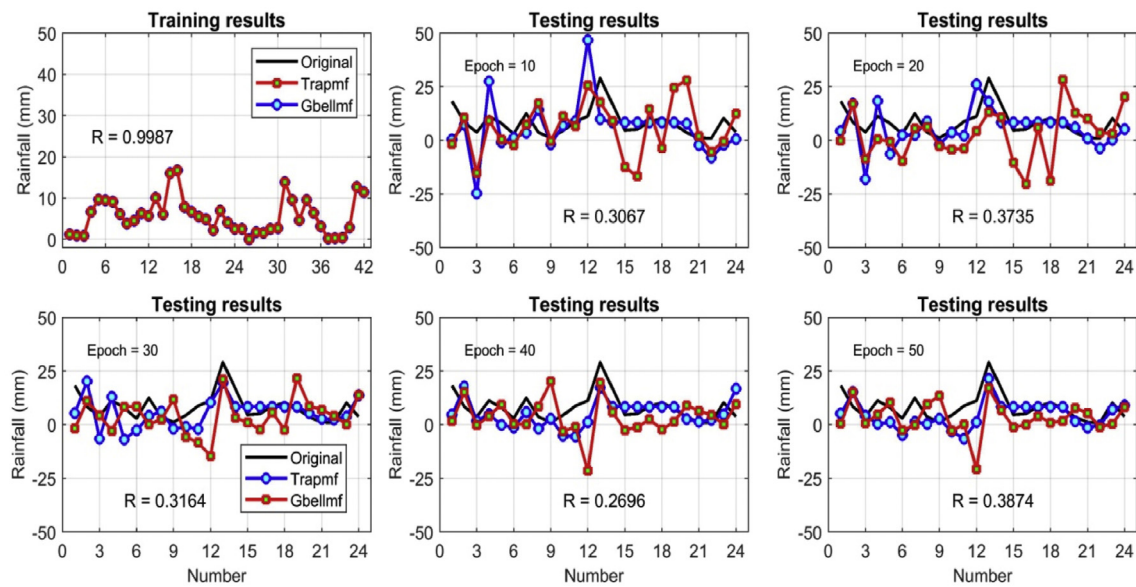


Fig. 6. Rainfall prediction results in South Tangerang with epoch (*t*) variations based on six-year data (2009–2014) where the comparison of data for training and testing used is 4:2.

Table 2

Summary performance for training and testing targets with epoch (*t*) variations where the comparison of training data and testing data are 4:2.

Performance Indicator	Generalized Membership Function (gbellmf)									
	<i>t</i> = 10	<i>t</i> = 20	<i>t</i> = 30	<i>t</i> = 40	<i>t</i> = 50	<i>t</i> = 60	<i>t</i> = 70	<i>t</i> = 80	<i>t</i> = 90	<i>t</i> = 100
MAPE Testing	1.467	1.628	1.298	1.275	1.185	1.159	1.134	1.113	1.094	1.075
MAPE Training	3.9×10^{-4}	1.6×10^{-4}	6.4×10^{-5}	3.7×10^{-5}	2.8×10^{-5}	2.6×10^{-5}	2.4×10^{-5}	2.2×10^{-5}	2.1×10^{-5}	2.0×10^{-5}
MSE Testing	0.125	0.126	0.125	0.125	0.125	0.125	0.125	0.125	0.125	0.125
MSE Training	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
Difference Test	0.352	0.391	0.311	0.306	0.284	0.278	0.272	0.267	0.262	0.258
RMSE	12.98	13.25	11.02	10.95	10.28	10.39	10.22	10.02	9.893	9.753

t = 100 s. The MSE for training and testing for 3:2 and 4:2 is consistently constant, both of which are independent on epoch variation. The lowest value obtained from MAPE training shows that the predicted value obtained is with a small error. A small error of training prediction value indicates that the data tested or validated will have a high probability of prediction. This will be fulfilled if the time series of long-term rainfall data is available completely so that it will minimize various forms of error in the prediction process, i.e. during the training and validation steps.

4. Conclusion

Using the ANFIS technique, monthly rainfall data in South Tangerang, Indonesia has been successfully predicted with an average performance of testing by about 80%. This technique is employing 6-year historical rainfall data to predict future rainfall where gbell mapping function at epoch 50s for the training and testing is 4:2 gives the best prediction results. Note that completeness of the data time series beforehand is very important for accurately predicting

rainfall or other parameters. In this ANFIS technique, time series containing not a number (NaN) or blank (no data) will drawback the ANFIS capability. Characteristics of data that are very volatile or very extreme will also produce low predictive results. Because 10-year historical rainfall data is not available completely throughout the year, future work will look at the rainfall prediction using other parameters that closely correlated with rainfall phenomena such as surface temperature, relative humidity, or wind speed. Further analysis will also look at what rainfall categories have the potential to trigger flash floods, especially urban flooding where rapid development of housing or the conversion of marginal areas into housing is of interest to be investigated.

Author statement

This paper is not published elsewhere or under review or submitted to another journal.

Conflicts of interest

The authors declare that there is no conflicts of interest.

Acknowledgements

This paper is supported by a TU001-2018 grant under the University of Malaya. The author highly appreciates the Meteorology, Climatology and Geophysics Agency (MCGA) of Indonesia or BMKG for providing surface meteorological data used in this study and LP2M Universitas Pembangunan Jaya where part of the work was initiated here and supported under the Internal Research Fund (001/PER-P2M/UPJ/05.19). Author gratitude is also extended to anonymous reviewer and journal editors for valuable and generous comments, which substantially contribute to improving the quality of the manuscript.

References

- [1] G.N. Alexander, Effect of catchment area on flood magnitude, *J. Hydrol.* 16 (1972) 225–240.
- [2] R. Ghazavi, A.M. Rabori, M.A. Reveshty, Modelling and assessment of urban flood hazards based on rainfall intensity-duration-frequency curves reformation, *Nat. Hazards Earth Syst. Sci. Discuss.* (2016), <https://doi.org/10.5194/nhess-2016-304>.
- [3] S.M. Mousavi, S. Roostaei, H. Rostamzadeh, Estimation of flood land use/land cover mapping by regional modelling of flood hazard at sub-basin level case study: Marand basin, *Geomatics, Nat. Hazards Risk* 10 (2019) 1155–1175.
- [4] M. Masoudian, S. Theobald, Influence of land surface topography on flood hydrograph, *J. Am. Sci.* 7 (2011) 354–361.
- [5] U.A. Ngamaliu-Nengoue, F.J. Martinez-Solano, P.L. Iglesias-Rey, D. Mora-Meliá, Multi-objective optimization for urban drainage or sewer networks rehabilitation through pipes substitution and storage tanks installation, *Water* 11 (2019) 935, <https://doi.org/10.3390/w11050935>.
- [6] Kompas on 26 October 2018. <https://megapolitan.kompas.com/read/2018/10/26/19180951/ini-titik-titik-rawan-banjir-di-tangerang-selatan> [Accessed on 9 February 2019].
- [7] C.M. Bhatt, G.S. Rao, Ganga floods of 2010 in Uttar Pradesh, north India: a perspective analysis using satellite remote sensing data, *Geomatics, Nat. Hazards Risk* (2014) 747–763, <https://doi.org/10.1080/19475705.2014.949877>.
- [8] P.C. Nayak, K.P. Sudheer, D.M. Rangan, K.S. Ramasastri, A neuro-fuzzy computing technique for modeling hydrological time series, *J. Hydrol.* 291 (1) (2004) 52–66.
- [9] F. Mekanik, M.A. Imteaz, A. Talei, Seasonal rainfall forecasting by adaptive network-based fuzzy inference system (ANFIS) using large scale climate signals, *Clim. Dynam.* 46 (2015) 3097–3111.
- [10] H. Hong, M. Panahi, A. Shirzadi, T. Ma, J. Liu, A.X. Zhu, W. Chen, I. Kougias, N. Kazakis, Flood susceptibility assessment in Hengfeng area coupling adaptive neuro-fuzzy inference system with genetic algorithm and differential evolution, *Sci. Total Environ.* 621 (2017) 1124–1141, <https://doi.org/10.1016/j.scitotenv.2017.10.114>.
- [11] D.T. Bui, K. Khosravi, S. Li, H. Shahabi, M. Panahi, V.P. Singh, K. Chapi, A. Shirzadi, S. Panahi, W. Chen, B. Ahmad, New hybrids of ANFIS with several optimization algorithms for flood susceptibility modeling, *Water* 10 (2018) 1210, <https://doi.org/10.3390/w10091210>.
- [12] W. Suparta, K.M. Alhasa, Modeling of Tropospheric Delays Using ANFIS, *International, Germany: Springer*, 2016. ISBN 978-3-319-28437-8.
- [13] W. Suparta, K.M. Alhasa, Precipitable water vapor using an adaptive neuro-fuzzy inference system in the absence of the GPS network, *J. Arch Meteorol. Climatol.* 55 (10) (2016) 2283–2300, <https://doi.org/10.1175/JAMC-D-15-0161.1>.
- [14] W. Suparta, K.M. Alhasa, A comparison of ANFIS and MLP models for the prediction of precipitable water vapor, in: 2013 IEEE international conference on IEEE space science and communication (IconSpace), 2013, pp. 243–248, <https://doi.org/10.1109/IconSpace.2013.6599473>.
- [15] W. Suparta, W.S. Putro, Comparison of tropical thunderstorm estimation between multiple linear regression, Dvorak, and ANFIS, *Bull. Electr. Eng. Inf.* 6 (2) (2017) 149–158, <https://doi.org/10.11591/eei.v6i2.648>.
- [16] M.P. Rajurkar, U.C. Kothiyari, U.C. Chaube, Artificial neural networks for daily rainfall–runoff modeling, *Hydrol. Sci. J.* 47 (6) (2002) 865–877, <https://doi.org/10.1080/02626660209492996>.
- [17] M. Sahin, R. Erol, A comparative study of neural networks and ANFIS for forecasting attendance rate of soccer games, *Math. Comput. Appl.* 22 (43) (2017), <https://doi.org/10.3390/mca22040043>.
- [18] J.S.R. Jang, ANFIS: adaptive network-based fuzzy inference systems, *IEEE Trans. Syst. Man Cybern.* 23 (1993) 665–685.
- [19] J.S. Jang, C.T. Sun, Neuro-fuzzy modeling and control, *Proc. IEEE* 83 (3) (1995) 378–406.
- [20] M. Huang, T. Zhang, J. Ruan, X. Chen, A new efficient hybrid intelligent model for biodegradation process of DMP with fuzzy wavelet neural networks, *Sci. Rep.* 7 (2017) 41239, <https://doi.org/10.1038/srep41239>.
- [21] Z. Liu, Chaotic time series analysis, *Math. Probl Eng.* (2010) 720190, <https://doi.org/10.1155/2010/720190>.
- [22] W. Wang, Y. Lu, Analysis of the mean absolute error (MAE) and the root mean square error (RMSE) in assessing rounding model, *IOP Conf. Ser. Mater. Sci. Eng.* 324 (2018), 012049, <https://doi.org/10.1088/1757-899X/324/1/012049>.
- [23] Mean absolute percentage error (MAPE), in: P.M. Swamidass (Ed.), *Encyclopedia of production and Manufacturing management*, Springer, Boston, MA, 2000, https://doi.org/10.1007/1-4020-0612-8_580.
- [24] N.G.N. Prasad, J.N.K. Rao, The estimation of the mean squared error of small-area estimators, *J. Am. Stat. Assoc.* 85 (409) (1990) 163–171, <https://doi.org/10.2307/2289539>.
- [25] B. Khoshnevisan, S. Rafiee, N. Omid, H. Mousazadeh, Development of an intelligent system based on ANFIS for predicting wheat grain yield on the basis of energy inputs, *Inf. Process. Agric.* 1 (1) (2014) 14–22, <https://doi.org/10.1016/j.inpa.2014.04.001>.
- [26] L.M. Brander, R.J.G.M. Florax, J.E. Vermaat, The empirics of wetland valuation: a comprehensive summary and a meta-analysis of the literature, *Environ. Resour. Econ.* 33 (2) (2006) 223–250.
- [27] V. Nourani, M. Komasi, A geomorphology-based ANFIS model for multi-station modeling of rainfall–runoff process, *J. Hydrol.* 490 (2013) 41–55.
- [28] A.A. Hancock, E.N. Bush, D. Stanicic, J.J. Kyncl, C.T. Lin, Data normalization before statistical analysis: keeping the horse before the cart, *Trends Pharmacol. Sci.* 9 (1) (1998) 29–32.



Wayan Suparta, PhD has experience as a lecturer, researcher, and high school teacher for over 25 years. His latest position is a lecturer at the Adisutjipto Institute of Aerospace Technology, Yogyakarta, Indonesia. Previously, a principal lecturer in the Department of Informatics, Universitas Pembangunan Jaya, Tangerang, Indonesia (6 Feb 2019 - 5 Feb 2020), a lecturer at the Department of Electrical Engineering, Sanata Dharma University, Yogyakarta, Adjunct Professor in the Department of Civil Engineering the Yogyakarta University of Technology. He is also a faculty member at the Doctor of Computer Science (DCS), Bina Nusantara University (BINUS) Jakarta and as a lecturer and researcher from the rank of Senior Lecturer to Associate Professor from July 2008 to April 2017 at the Space Science Centre (ANGKASA) of Universiti Kebangsaan Malaysia (UKM). He began his career as a teacher of Physics, Electronics, and Computers in high school at SMUK Cor Jesu Malang (1994–1997) and a teacher at SMUK Santo Aloysius Bandung (1997–2000). Then became a Lecturer in Electrical Engineering at College Legenda Group, Malaysia (2000 - 2004). After graduating from the Faculty of Engineering, UKM in March 2008, he was appointed as a Post Doctoral Fellow (2007–2008), subsequently appointed as a Senior Lecturer (1 July 2008 - 3 April 2012) and Associate Professor (4 April 2012 - 3 April 2017). His teaching area is in electronics/electrical engineering, informatics, physics, and statistics, while his research is focused on GPS/GNSS Technology, Communication Systems (Remote Sensing), Smart Systems, Artificial Neural Network Applications, and Data Mining.